

## HARRELL FIELD, AR, USA

WMO: 723425

Lat: 33.617N

Lon: 92.767W

Elev: 132

StdP: 14.63

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00373

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 20.2       | 24.7 | 6.6                           | 7.7 | 29.2 | 11.6 | 10.0 | 33.4 | 17.8                  | 60.3 | 16.3 | 56.7 | 1.7                   | 10   | 0.382 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.3                   | 99.1            | 76.1 | 96.7 | 75.9 | 94.5 | 75.9 | 79.4                | 92.1 | 78.8 | 90.9 | 78.0 | 89.6 | 5.4                  | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 76.8                            | 140.4 | 82.7 | 75.3 | 133.3 | 82.3 | 74.9 | 131.4 | 82.1 | 43.1          | 91.0 | 42.2 | 91.4 | 41.4 | 90.9 | 82.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |       |
| 15.9              | 12.9 | 11.6 | DB                         | 13.4 | 101.8              | 5.4 | 2.2                                                | 9.5 | 103.4      | 6.4 | 104.6      | 3.3 | 105.8      | -0.6 | 107.4 |
|                   |      |      | WB                         | 12.4 | 81.2               | 5.5 | 0.7                                                | 8.5 | 81.7       | 5.3 | 82.1       | 2.2 | 82.5       | -1.7 | 83.0  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 63.6        | 42.6  | 47.4  | 55.6  | 63.6  | 71.8  | 79.9  | 81.5  | 81.3  | 76.2  | 63.5  | 52.7  | 46.5  | (6)  |
|                                                                        | DBStd       | 15.89       | 10.24 | 10.46 | 9.60  | 7.19  | 6.30  | 3.79  | 4.20  | 4.54  | 6.16  | 8.56  | 9.06  | 9.34  | (7)  |
|                                                                        | HDD50       | 758         | 276   | 162   | 52    | 4     | 0     | 0     | 0     | 0     | 0     | 5     | 74    | 185   | (8)  |
|                                                                        | HDD65       | 2732        | 696   | 499   | 317   | 109   | 20    | 0     | 0     | 0     | 2     | 133   | 379   | 577   | (9)  |
|                                                                        | CDD50       | 5736        | 47    | 88    | 225   | 413   | 677   | 897   | 976   | 971   | 785   | 424   | 156   | 77    | (10) |
|                                                                        | CDD65       | 2235        | 3     | 5     | 25    | 68    | 231   | 447   | 511   | 506   | 337   | 88    | 11    | 3     | (11) |
|                                                                        | CDH74       | 23428       | 5     | 35    | 248   | 688   | 2162  | 4845  | 5582  | 5510  | 3297  | 946   | 93    | 17    | (12) |
|                                                                        | CDH80       | 11461       | 0     | 3     | 41    | 133   | 837   | 2508  | 2952  | 2963  | 1654  | 353   | 17    | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 4.0         | 4.4   | 4.9   | 5.4   | 5.0   | 4.2   | 3.8   | 3.2   | 3.2   | 2.9   | 3.2   | 3.7   | 4.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 54.3        | 4.6   | 4.3   | 5.0   | 5.0   | 5.4   | 3.8   | 3.8   | 2.8   | 3.8   | 5.3   | 4.0   | 5.2   | (15) |
|                                                                        | PrecMax     | 83.1        | 9.3   | 10.0  | 11.9  | 16.7  | 12.9  | 8.1   | 10.4  | 10.0  | 10.5  | 21.5  | 10.0  | 8.8   | (16) |
|                                                                        | PrecMin     | 32.4        | 0.2   | 0.5   | 1.9   | 1.3   | 1.7   | 1.5   | 0.5   | 0.4   | 1.2   | 0.1   | 0.4   | 0.6   | (17) |
|                                                                        | PrecStd     | 12.5        | 2.2   | 2.4   | 2.6   | 3.7   | 3.1   | 2.0   | 2.8   | 2.0   | 2.5   | 4.5   | 2.5   | 2.1   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 73.5  | 78.9  | 83.9  | 86.0  | 92.9  | 99.4  | 100.0 | 101.9 | 99.5  | 93.1  | 82.1  | 76.6  | (19) |
|                                                                        |             | MCWB        | 59.2  | 64.4  | 65.7  | 67.0  | 72.7  | 74.6  | 77.0  | 77.7  | 74.5  | 71.5  | 67.4  | 66.2  | (20) |
|                                                                        | 2%          | DB          | 70.0  | 73.2  | 81.0  | 83.5  | 90.4  | 95.4  | 98.7  | 99.2  | 96.8  | 88.3  | 75.4  | 72.4  | (21) |
|                                                                        |             | MCWB        | 58.6  | 62.3  | 64.1  | 67.8  | 72.1  | 75.8  | 76.8  | 77.0  | 73.9  | 71.8  | 63.1  | 63.5  | (22) |
|                                                                        | 5%          | DB          | 64.5  | 70.1  | 76.6  | 81.4  | 88.0  | 93.2  | 95.1  | 95.4  | 93.1  | 83.8  | 72.4  | 67.9  | (23) |
|                                                                        |             | MCWB        | 56.6  | 60.9  | 61.9  | 66.3  | 72.0  | 75.3  | 76.9  | 76.6  | 74.0  | 70.3  | 62.7  | 61.4  | (24) |
|                                                                        | 10%         | DB          | 61.2  | 65.8  | 72.4  | 78.6  | 84.4  | 91.2  | 93.0  | 93.1  | 90.3  | 79.3  | 68.2  | 63.8  | (25) |
|                                                                        |             | MCWB        | 52.6  | 58.2  | 60.2  | 64.5  | 70.8  | 75.4  | 76.4  | 76.5  | 73.5  | 66.9  | 60.6  | 58.0  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 65.7  | 67.7  | 68.7  | 73.2  | 77.3  | 79.7  | 80.3  | 80.9  | 78.9  | 76.7  | 70.7  | 68.4  | (27) |
|                                                                        |             | MCDB        | 71.5  | 73.9  | 78.2  | 80.4  | 85.9  | 90.9  | 91.9  | 92.7  | 89.4  | 86.0  | 78.4  | 73.9  | (28) |
|                                                                        | 2%          | WB          | 62.5  | 65.7  | 66.6  | 70.8  | 75.3  | 78.6  | 79.3  | 79.7  | 77.5  | 74.7  | 67.9  | 66.0  | (29) |
|                                                                        |             | MCDB        | 66.7  | 71.8  | 75.8  | 79.2  | 84.6  | 90.0  | 92.1  | 92.7  | 88.3  | 83.8  | 72.9  | 70.5  | (30) |
|                                                                        | 5%          | WB          | 58.9  | 62.7  | 64.7  | 69.2  | 73.9  | 77.7  | 78.5  | 78.9  | 76.4  | 72.7  | 65.2  | 63.2  | (31) |
|                                                                        |             | MCDB        | 63.0  | 68.0  | 73.4  | 77.4  | 84.0  | 88.9  | 90.8  | 91.5  | 87.0  | 80.3  | 69.8  | 66.4  | (32) |
|                                                                        | 10%         | WB          | 53.8  | 59.3  | 62.4  | 67.3  | 72.5  | 76.9  | 77.8  | 78.0  | 75.2  | 70.1  | 61.9  | 58.8  | (33) |
|                                                                        |             | MCDB        | 59.9  | 65.1  | 69.7  | 75.1  | 82.3  | 87.7  | 89.5  | 89.7  | 85.3  | 76.1  | 67.2  | 63.0  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 22.2  | 22.1  | 24.5  | 24.1  | 22.5  | 22.5  | 21.3  | 22.4  | 23.5  | 25.4  | 24.7  | 21.2  | (35) |
|                                                                        |             | MCDBR       | 29.0  | 26.7  | 27.3  | 26.3  | 25.8  | 24.7  | 24.1  | 25.5  | 28.2  | 28.4  | 27.7  | 26.6  | (36) |
|                                                                        | 5% WB       | MCWBR       | 18.9  | 16.9  | 13.8  | 12.7  | 10.0  | 8.3   | 7.4   | 8.0   | 9.5   | 13.0  | 16.4  | 18.3  | (37) |
|                                                                        |             | MCWBR       | 20.9  | 21.8  | 22.6  | 21.4  | 20.9  | 21.5  | 21.2  | 22.4  | 21.7  | 23.1  | 22.1  | 21.8  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.328 | 0.337 | 0.365 | 0.397 | 0.439 | 0.456 | 0.487 | 0.480 | 0.418 | 0.365 | 0.344 | 0.323 | (40) |
|                                                                        |             | taud        | 2.506 | 2.480 | 2.419 | 2.328 | 2.268 | 2.258 | 2.199 | 2.208 | 2.359 | 2.480 | 2.491 | 2.529 | (41) |
|                                                                        | Ebn at Noon |             | 281   | 290   | 289   | 283   | 271   | 265   | 256   | 256   | 268   | 276   | 272   | 276   | (42) |
|                                                                        |             | Edh at Noon | 27    | 30    | 35    | 40    | 43    | 43    | 46    | 45    | 37    | 30    | 27    | 25    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 818   | 937   | 1334  | 1662  | 1831  | 2023  | 1972  | 1830  | 1569  | 1252  | 895   | 718   | (44) |
|                                                                        | RadStd      |             | 78    | 154   | 115   | 109   | 146   | 163   | 124   | 152   | 138   | 170   | 104   | 83    | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (24 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page